

Bidirectional Heuristic Search on 3D Voxel Maps

Gilad Shpigelman

[partner name]

Ben-Gurion University of the Negev

Search Methods in Artificial Intelligence (237-2-5513) — Advisor: Lior Siag

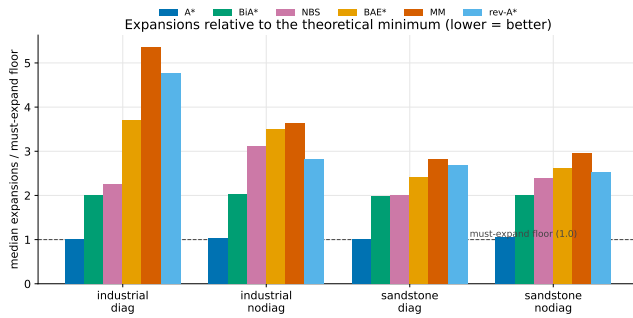


Figure 1: Median expansions relative to the must-expand floor, per algorithm across domains and movement models. Lower is better; 1.0 is optimal.

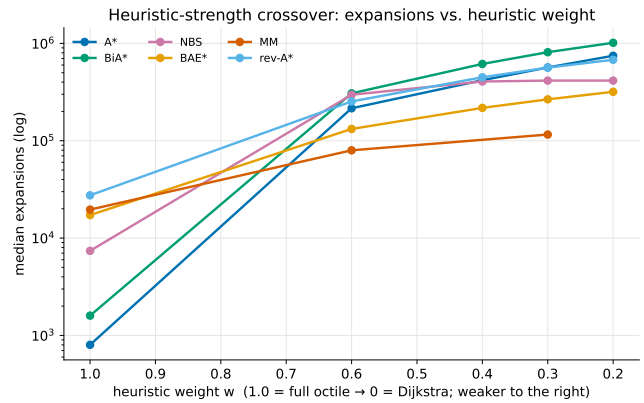


Figure 2: Expansions vs. heuristic weight w ($h' = w h$). A^* is best at full strength but is overtaken by MM/BAE* as the heuristic weakens.

Abstract

1 Introduction and Literature Review

1.1 Problem and motivation

1.2 Related work

2 Methodology

2.1 Domain and movement model

2.2 System: HOG2, the loader, and the algorithms

2.3 Metric: the must-expand floor

2.4 Correctness

2.5 Experimental setup (reproducibility)

3 Experimental Results

3.1 Expansions vs. the theoretical floor

3.2 Who actually wins per instance

3.3 Directional asymmetry

3.4 The heuristic-strength crossover

3.5 Descent (maze-like domain)

3.6 GBFS trade-off

4 Conclusions and Summary

1 Code and Reproducibility

Code, data, and figures: <https://github.com/gshpigelman/bidirectional-heuristic-search>

final-project-voxel-bidirectional-search.

References

- [1] Author, A. *Copy the entries from background-and-plan.tex*. 2025.